



A test for the redshift dependence of σ_8 using $f\sigma_8$ measurements

Siddhant Manna^a, Shantanu Desai^b

Department of Physics, IIT Hyderabad, Kandi, Telangana 502284, India

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Abstract We search for a redshift dependence of σ_8^0 using 23 growth rate measurements from redshift space distortions and peculiar velocity measurements as a consistency check of Λ CDM. For this purpose we use the dataset from Sagredo et al. (Phys Rev D 98:083543, 2018) consisting of 22 measurements, which has been vetted for internal consistencies. Eighteen of these measurements have been obtained from the Gold-2017 sample, collated from multiple redshift space distortion surveys between 2009 and 2016, whereas the remaining four have been obtained from the eBOSS DR14 quasar survey. We also added one additional data point from the BOSS DR12 CMASS galaxy sample. We find that for this dataset the σ_8^0 values are consistent between the low redshift and high redshift samples using three different redshift cuts. This implies that the growth rate measurements are consistent with a constant σ_8^0 in accord with Λ CDM, assuming there are no uncontrolled systematics in this dataset.

1 Introduction

The current concordance model (Λ CDM) consisting of 70% dark energy and 25% cold dark matter is the prevailing cosmological model for the universe. It is based on the Friedmann–Lemaître–Robertson–Walker (FLRW) metric and assumes that the universe is spatially flat, homogeneous, and isotropic [1]. The model has been very successful in explaining a wide range of cosmological observations, including the cosmic microwave background (CMB), the abundance of light elements, and the large-scale structure of the universe [2].

Despite the overall success of the concordance Λ CDM model of cosmology, over the last few years, there are still a few data-driven tensions with observations. The most vexing

problem is the intriguing discrepancy between the Hubble constant (H_0) values obtained from local measurements and those inferred from CMB data. Local measurements of H_0 , primarily obtained from observing Cepheid variable stars and supernovae of Type Ia, typically yield a higher value, around 73 km/s/Mpc compared to the CMB inferred value of around 67 km/s/Mpc. This discrepancy, known as the Hubble tension, poses a challenge to the Λ CDM model and has spurred a search for potential explanations [3–11]. Another vexing problem is the σ_8/S_8 tension. Here, σ_8 is a measure of the variance of the matter density fluctuations on a scale of $8 h^{-1}$ Mpc and at redshift $z = 0$ [12]. For the rest of the manuscript we use σ_8^0 to refer to σ_8 inferred at redshift of 0 to distinguish between growth rate measurements, which depend on σ_8 inferred at non-zero redshifts. One can then define a weighted amplitude of matter fluctuations (S_8) by combining it with matter density (Ω_m) and σ_8^0 as follows:

$$S_8 \equiv \sigma_8^0 \sqrt{\frac{\Omega_m}{0.3}}. \quad (1)$$

Recent measurements of S_8 from different cosmological probes such as CMB and galaxy survey measurements have shown that S_8 inferred from Planck has a higher value of S_8 as compared to that obtained from galaxy surveys [2, 13–21]. This is known as the S_8 tension. An up-to-date summary of all recent tensions of the standard Cosmological model with observations is reviewed in [13, 22–24].

Concurrent with these tensions, there have been recent results, which have shown that some of the above cosmological constants depend on the redshift of the probes used to determine these constants. In other words they have been found to be redshift-dependent. For instance, some works have suggested that the Hubble constant might not be constant but could instead evolve with redshift, decreasing with increasing redshift [25–33]. In a similar vein, there have been claims that the matter density parameter Ω_m , increases with effective redshift [25, 29, 34–41].

^a e-mail: ph22resch11006@iith.ac.in

^b e-mail: shntn05@gmail.com (corresponding author)

Some of these works have also found an anti-correlation between the variation of Ω_m and H_0 . Such a variation of the above cosmological constants with redshift implies a breakdown of the Λ CDM model assuming there are no uncontrolled systematic errors, since σ_8^0 and H_0 are integration constants within the Λ CDM model for an FLRW metric and do not evolve with redshift by definition [42]. Therefore, if their inferred values using probes at different redshifts vary, it would signal a breakdown of Λ CDM.

If Ω_m also varies as a function of redshift, this implies that the observed increase in S_8 must be driven by an increase in σ_8^0 . Therefore, in order to test this *ansatz*, [43] (A23, hereafter) searched for a redshift dependence of S_8 within Λ CDM and assuming a constant Ω_m . For this purpose, they used $f\sigma_8$ measurements from peculiar velocity and redshift space distortion measurements, where $f = \frac{d \ln \delta}{d \ln a}$ [44]. This work checked for a redshift dependence of S_8 after bifurcating the dataset into two samples using the redshift cut of $z = 0.7$. These measurements are agnostic to galaxy bias and hence provide a more robust probe. Using these measurements, A23 showed that S_8 increases with redshift.

In this work, we carry out a slight variant of the analysis implemented in A23 using an independent dataset, where we search for a redshift dependent σ_8^0 by dividing the samples according to redshift using three redshift cuts. Since by definition, σ_8^0 should not depend on the redshift of the probes used for its measurement, a statistically significant difference between its estimated value between the different redshift probes would provide a hint of breakdown of Λ CDM or point to some unknown systematics in the data. This manuscript is structured as follows. In Sect. 2, we summarize the results of A23. In Sect. 3 we describe the data used, along with the analysis. Finally, our conclusions can be found in Sect. 4. For our analysis, we assume a flat Λ CDM cosmology with $h = 0.7$.

2 Summary of A23

A23 performed a consistency test of the Planck- Λ CDM cosmology to investigate the redshift evolution of the parameter S_8 . They first subjected the analysis to 20 growth rate data points from peculiar velocity and redshift-space distortion (RSD) data [45], and as a cross-check also redid their analysis with 66 growth rate data points obtained from [46] for a more comprehensive analysis. A23 imposed a prior on $\Omega_m = 0.3111 \pm 0.0056$, which is informed by observational data from both the Planck CMB and BAO experiments [2]. In their study, they employed a generalized matter density parameter, denoted by $\Omega(z)$ [42]. It quantifies the relative density of matter in the universe compared to a critical density. It is defined according to:

$$\Omega(z) := \frac{\Omega_m(z)}{H(z)^2/H_0^2} = \frac{\Omega_m(1+z)^3}{1 - \Omega_m + \Omega_m(1+z)^3}. \quad (2)$$

The corresponding equation for $f\sigma_8(z)$ can be expressed as:

$$f\sigma_8(z) = \sigma_8^0 \Omega^{6/11}(z) \exp\left(-\int_0^z \frac{\Omega^{6/11}(z')}{1+z'} dz'\right). \quad (3)$$

The expression in Eq. 3 is a valid approximation for Λ CDM [42] and the maximum difference with the exact expression (involving hypergeometric functions) [44] is about 1% at $z = 0$ [43]. Since our aim is to test for a redshift dependence, we use the aforementioned equation, which would be accurate enough for our purpose. A Bayesian regression analysis using three free parameters (Ω_m , σ_8 , and S_8) was carried out using Eq. 3 after bifurcating the $f\sigma_8$ measurements into a low redshift and high redshift sample with the boundary at $z = 0.7$. Using 20 measurements, A23 found that for the low redshift data, Ω_m and σ_8 were consistent with Planck 2020, but S_8 was discrepant at about 3.4σ . For the high redshift data, the tension with the Planck 2020 measurements for all the three parameters is between $1 - 2\sigma$. For the 66 growth measurements, the data was bifurcated at $z = 1.1$. The S_8 values between low and high redshifts differ by about 2.8σ . Also, the S_8 value below $z = 1.1$ is in tension with the Planck value at about 5σ .

3 Data selection and analysis

The dataset in A23 was based on the compilations in Refs. [46,47]. However, the measurements of $f\sigma_8(z)$ can be prone to various systematic effects, necessitating careful consideration when selecting the sample. Since the datasets used in A23 could exhibit correlations among themselves, we use the dataset in [48] which consists of 22 growth rate measurements (cf. Table I of [48]). This dataset has been extensively scrutinized for systematics and internal consistency checks using the Bayesian approach prescribed in [49]. Eighteen data points in this sample are based on the Gold-2017 compilation, consisting of independent measurements of $f\sigma_8$ collated from different redshift space distortion based surveys between 2009 and 2016 [44]. These surveys include 6dFGS, IRAS, 2MASS, SDSS, GAMA, SDSS-LRG-200, SDSS-CMASS, BOSS-LOWZ, WiggleZ, Vipers, FastSound. These measurements depend on the fiducial cosmology assumed by the surveys for the analysis. These were corrected by rescaling the growth rate measurements by the ratio of $H(z)D_a(z)$ of the cosmology to the fiducial one. More details on this rescaling can be found in [44]. The remaining four measurements were obtained from [50], obtained from SDSS-IV eBOSS DR14 quasar survey [51]. To this dataset, we added a recent measurement of $f\sigma_8 = 0.462 \pm 0.020$

Fig. 1 Marginalized credible intervals for Ω_m and σ_8 using 23 growth rate data points. The $f\sigma_8(z)$ data is bifurcated at $z = 0.7$. The innermost contour represents a 68% credible interval, while the outermost contour corresponds to a 95% credible interval

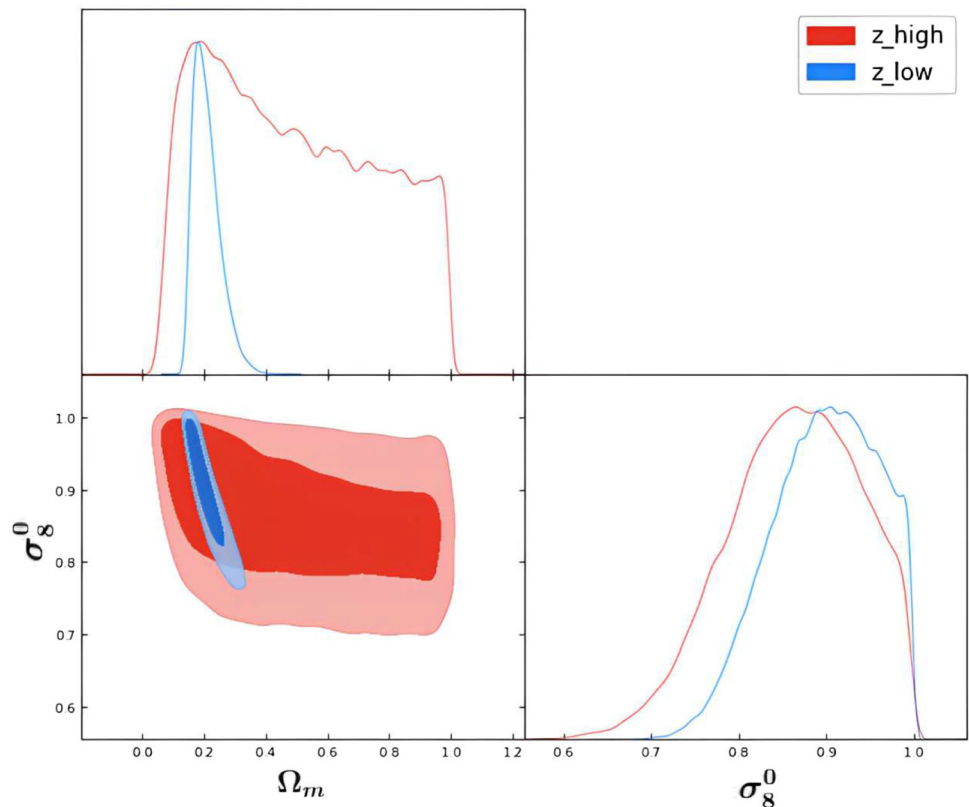


Table 1 Priors on the cosmological parameters used for our analyses

Parameter	Distribution	Prior
Ω_m	Uniform	$\mathcal{U}(0, 1.0)$
σ_8^0	Uniform	$\mathcal{U}(0.7, 0.9)$

Table 2 Best-fit values for σ_8^0 and Ω_m for all the four analyses carried out in this work. The corresponding credible intervals can be found in Fig. 1, Fig. 2, and Fig. 3

Data points	Ω_m	σ_8^0	Redshift cut
07	0.489 ± 0.26	0.863 ± 0.075	$z \geq 0.7$
16	0.21 ± 0.044	0.898 ± 0.059	$z < 0.7$
12	0.387 ± 0.22	0.841 ± 0.078	$z \geq 0.4$
11	0.278 ± 0.096	0.803 ± 0.105	$z < 0.4$
16	0.206 ± 0.065	0.906 ± 0.061	$z \geq 0.2$
07	0.376 ± 0.183	0.723 ± 0.154	$z < 0.2$

at $z \approx 0.525$ obtained from clustering analysis of the BOSS DR12 CMASS galaxy sample [52].

To investigate the redshift dependence, instead of carrying out regression using three free parameters (σ_8^0 , S_8 and Ω_m) as in A23, we use only two parameters, σ_8^0 and Ω_m . We then

use a Gaussian likelihood defined as follows:

$$\ln \mathcal{L}(\Omega_m) = -\frac{1}{2} \sum_i \frac{[f\sigma_{8,z_i} - \hat{f}\sigma_{8,z_i}]^2}{\sigma_{f\sigma_{8,z_i}}^2} \quad (4)$$

where $f\sigma_8$ is defined in Eq. 3. We mostly use similar priors as in A23. The only exception is Ω_m , where we used a uniform prior $\in [0, 1]$. For σ_8^0 , we use a uniform prior. A tabular summary of the parameters and their respective priors can be found in Table 1. We then sample the above likelihood using the emcee MCMC sampler [53]. The marginalized credible intervals are obtained using the `getdist` package [54].

We carried out two different analyses. We first did a Bayesian regression on the above parameters after bifurcating the dataset into two redshift samples with cuts at $z = 0.2$, $z = 0.4$, and $z = 0.7$.

The best-fit parameters for Ω_m and σ_8 for 23 can be found in Table 2. The corresponding marginalized contours for both the low and high redshift samples can be found in Fig. 1.

We summarize our results as follows:

- The difference in Ω_m values between both the low redshift and high redshift sample for both sets of data is less than 1σ , for all the three redshift cuts. Therefore, we do not find any evidence for a redshift dependence of Ω_m .
- The difference in σ_8 values between the low redshift and high redshift sample is also less than 1σ for our 23 measurements dataset for all the three redshift cuts.

Fig. 2 Same as Fig. 1 but $f\sigma_8(z)$ data is bifurcated at $z = 0.4$

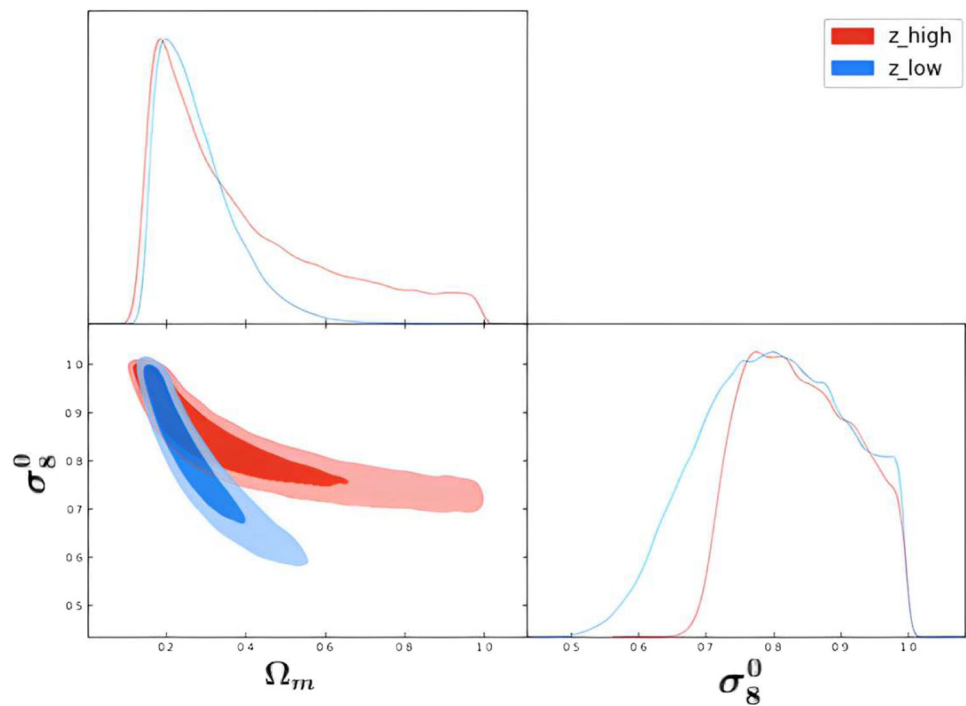
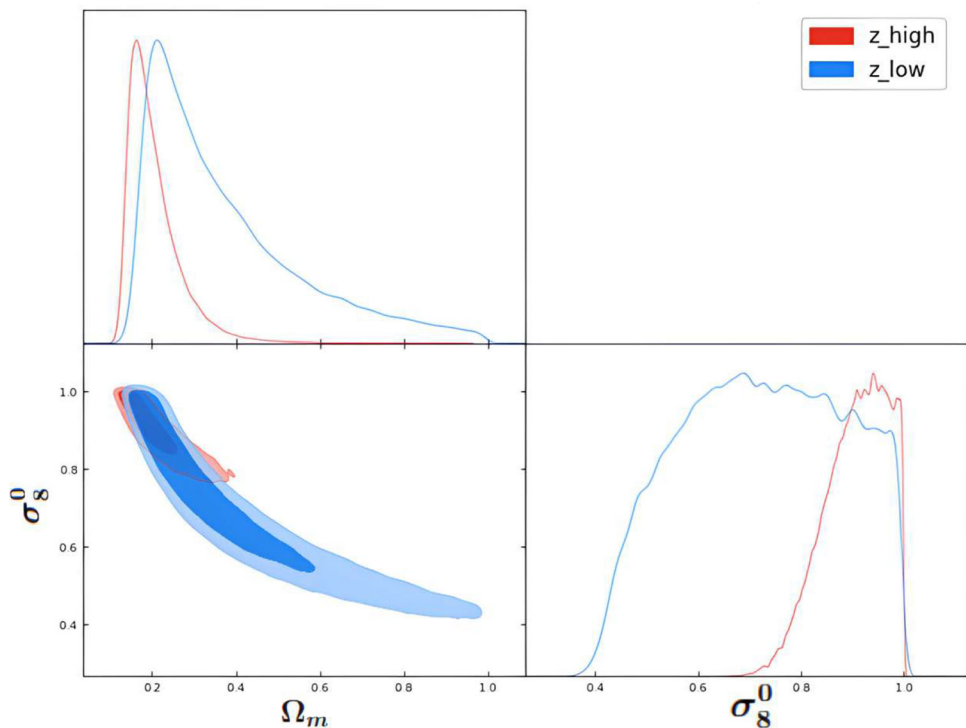


Fig. 3 Same as Fig. 1 but $f\sigma_8(z)$ data is bifurcated at $z = 0.2$



Therefore, unlike A23, we do not find any dependence of σ_8^0 with redshift, and our results are in accord with Λ CDM.

4 Conclusions

To get some insight on the σ_8/S_8 tension problem, A23 looked for a redshift dependence of σ_8^0 and S_8 using $f\sigma_8$ measurements obtained from peculiar velocity and RSD measure-

ments. They carried out a regression analysis using Ω_m , σ_8^0 , and S_8 and found a redshift dependence of S_8 with a discrepancy of $1.6 - 2.8\sigma$ between the low and high redshift samples.

We carry out a variant of the above analysis using an independent dataset which has been thoroughly vetted using internal consistency checks [48], by doing a regression analysis using only Ω_m and σ_8^0 , and bifurcating the dataset into low and high redshift samples, after imposing a redshift cut of

$z = 0.2$, $z = 0.4$, and $z = 0.7$. We use a uniform prior on Ω_m and uniform prior on σ_8^0 . We find that σ_8^0 values are consistent between the low redshift and high redshift samples to within 1σ . Therefore, we do not find evidence for an increase of σ_8^0 with redshift which was found in A23, without incorporating S_8 in our analysis. This implies that there is no breakdown of Λ CDM using our analysis, assuming there are no uncontrolled systematics in the dataset hitherto analyzed.

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Code Availability Statement Code/software will be made available on reasonable request. [Author's comment: The code/software generated during and/or analysed during the current study is available from the corresponding author on reasonable request.]

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